

Design and simulation of BLDC motor control using simulink-MATLAB: A detailed approach

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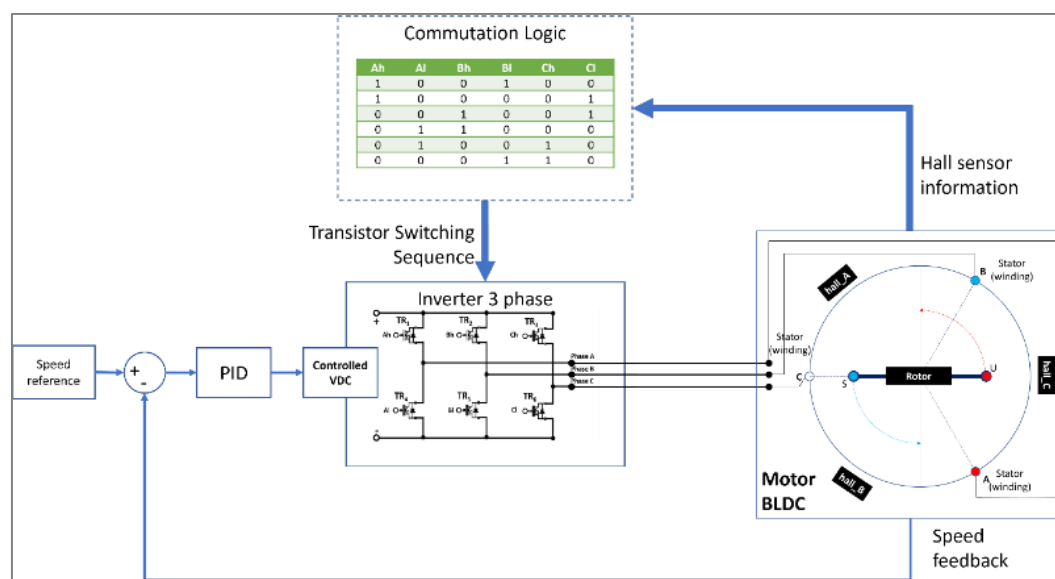
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Highlights:

- Comprehensive BLDC control simulation in MATLAB/Simulink, accurately reproducing Hall sensor signals, back-EMF waveforms, switching sequences, and motor dynamic responses.
- Open-loop validation of Six-Step Commutation, confirming correct switching logic and enabling clear analysis of commutation behavior prior to closed-loop implementation.
- Closed-loop PID-based speed regulation with automatic tuning, maintaining the reference speed under varying commands and load conditions with deviations limited to $\pm 2\%$ and short, well-damped transients.
- Insight into phase current and electromagnetic torque characteristics, showing proportional scaling with speed and load, and providing guidance for practical enhancements such as anti-windup strategies, ripple mitigation, and optimized commutation timing.

Abstract

Brushless Direct Current (BLDC) motors are increasingly utilized across various applications due to their high efficiency and reliability. However, their control requires precise handling, especially during the commutation process. This study presents a detailed simulation design for BLDC motor control using MATLAB Simulink, focusing on the Six-Step Commutation method and PID-based speed regulation. The methodology involves constructing an open-loop model to analyze commutation behavior, followed by a closed-loop system using PID controllers with automatic parameter tuning. The simulation demonstrates accurate replication of hall sensor signals, back-EMF waveforms, switching patterns, and motor responses. Results reveal that the PID controller effectively maintains target speed across varying reference inputs and load conditions, while phase

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current and electromagnetic torque increase proportionally with speed and load. Results confirm correct switching in open loop and show that, in closed loop, the controller maintains speed within $\pm 2\%$ of the target with brief, well-damped transients. Phase current and torque responses scale with speed and load, informing practical refinements (anti-windup, ripple mitigation, soft-commutation timing). The findings certify that simulation is a vital step to ensure functional logic and hardware readiness, minimizing risks and enhancing system performance prior to physical implementation.

Keywords: BLDC motor; Six-Step Commutation; PID control; MATLAB Simulink; Speed control simulation

1. Introduction

Electric motors convert electrical energy into mechanical motion, performing the inverse of electric generators that produce electrical energy from mechanical input. They are used extensively in modern applications ranging from toys and household appliances to industrial machinery and electric vehicles [1]. An electric motor typically consists of a rotor and stator; rotation is produced by the interaction between magnetic fields generated by windings or permanent magnets [2]. Depending on the configuration, either the rotor or the stator may contain the permanent magnets, and motors are broadly categorized as DC and AC types [3]. The friction and wear associated with the brushes in conventional DC motors have led to the development of the more efficient Brushless DC (BLDC) motor [4], which replaces mechanical commutation with an electronically controlled system.

The implementation of BLDC motors has been widely studied and applied in various systems, including water pumps, electric bicycles, and electric vehicles [1], [5], [6], [7]. Two main types of BLDC motors exist, distinguished by their back electromotive force (back-EMF) waveforms: trapezoidal and sinusoidal. Back-EMF is the voltage induced in the windings due to changes in current and rotor position.

Two DC brushless motor categories are commonly distinguished by their back-EMF waveform: trapezoidal and sinusoidal. In this paper, we use the term BLDC strictly for trapezoidal back-EMF drives, whereas sinusoidal back-EMF machines are referred to as PMSM. This naming is kept consistent throughout the manuscript. BLDC motors characterized by sinusoidal back-EMF are more generally designated as Permanent Magnet Synchronous Motors (PMSM) [8], [9], [10]. In this article, the term BLDC especially denotes motors characterized by trapezoidal back-EMF.

The deployment of BLDC motors necessitates meticulous planning, as their rotor rotation relies on accurate six-step electronic commutation controlled by Hall sensor feedback. Unlike brushed DC motors, which achieve mechanical commutation through brushes, BLDC drives depend on electronic switching sequences synchronized with rotor position. Incorrect commutation sequences may prevent startup or impose electrical and thermal stress on inverter switches, highlighting the importance of precise design and control implementation. A prevalent and effective technique for operating BLDC motors is the Six-Step Commutation method [11].

Moreover, the integration of electric motors, such as BLDCs, is frequently linked to the regulation of rotational velocity and the management of load during operational processes. Among the numerous strategies accessible, Proportional-Integral-Derivative (PID) control is one of the most extensively utilized methods. PID controllers can be configured with solely a proportional term (P), a combination of proportional and integral terms (PI), or all three components (PID), contingent upon the required system response and complexity [12], [13].

For this reason, the understanding of controlling BLDC motors must be well known. This paper attempts to explain in detail how to build a BLDC motor control simulation using MATLAB software. The sections that will be discussed in this paper include introduction, proposed simulation method, simulation results and discussion and ends with a conclusion. The introduction includes the background of writing the paper, some theoretical basis of BLDC which includes how BLDC can rotate, what circuit is used to rotate it, what methods are used in the circuit and the basis of the PID control method in BLDC and an explanation related to MATLAB and Simulink software used as a simulation tool.

1.1. Basic Operating Principle of BLDC Motors

Figure 1 illustrates the rotor rotation process of a BLDC motor. The functioning is considerably more intricate than that of a standard DC motor, requiring a comprehensive grasp of commutation

logic for effective and dependable control. Erroneous commutation sequences can lead to both motor failure to initiate and potential hardware damage. To alleviate such risks and verify control logic, simulation is a crucial phase in system development [14]. This study seeks to deliver a thorough examination of the design and simulation of a Six-Step Commutation technique for BLDC motors, alongside the execution of PID-based speed control utilizing MATLAB/Simulink software.

In a BLDC motor, the rotor generally comprises a permanent magnet, whilst the stator consists of three-phase windings. The essential operating principle of a BLDC motor entails the accurate "commutation" or sequencing of voltages delivered to the stator windings, so generating an electromagnetic field that alternately attracts and repels the rotor's magnetic poles. This alternating connection generates rotational motion. Voltage regulation for each stator phase is accomplished by a three-phase inverter, which methodically energizes the stator windings to produce a revolving magnetic field. Precise commutation necessitates real-time rotor position feedback, typically supplied by integrated Hall effect sensors.

Figure 1 depicts the rotational mechanism of a BLDC motor rotor with a pair of magnetic poles (north – U and south – S), three stator windings identified as A, B, and C, and three Hall sensors designated as hall_A, hall_B, and hall_C. In **Figure 1a**, a positive voltage ($+V_{dc}$) is applied to phase A, while a negative voltage ($-V_{dc}$) is applied to phase B. This generates an electromagnetic field directed from A to B (as indicated by the central axis of the circle), prompting the rotor to align correspondingly: the U (north) pole is attracted to B, while the S (south) pole is drawn to A. At this location, the Hall sensors hall_C and hall_A register the presence of the U pole, however hall_B does not.

In the succeeding condition illustrated in **Figure 1b**, a voltage of $+V_{dc}$ is supplied to phase A, while $-V_{dc}$ is applied to phase C. The resultant electromagnetic field develops between phases A and C, causing the U pole to shift towards C and the S pole towards A. The U pole is solely identified by hall_A in this condition. The commutation sequence progresses cyclically through the configurations depicted in **Figure 1c** to **Figure 1f**, ultimately reverting to the original state illustrated in **Figure 1a**. These six commutation processes comprise a complete electrical rotation. The comprehensive commutation logic associated with the Hall sensor readings and phase switching sequence is defined in **Table 1**.

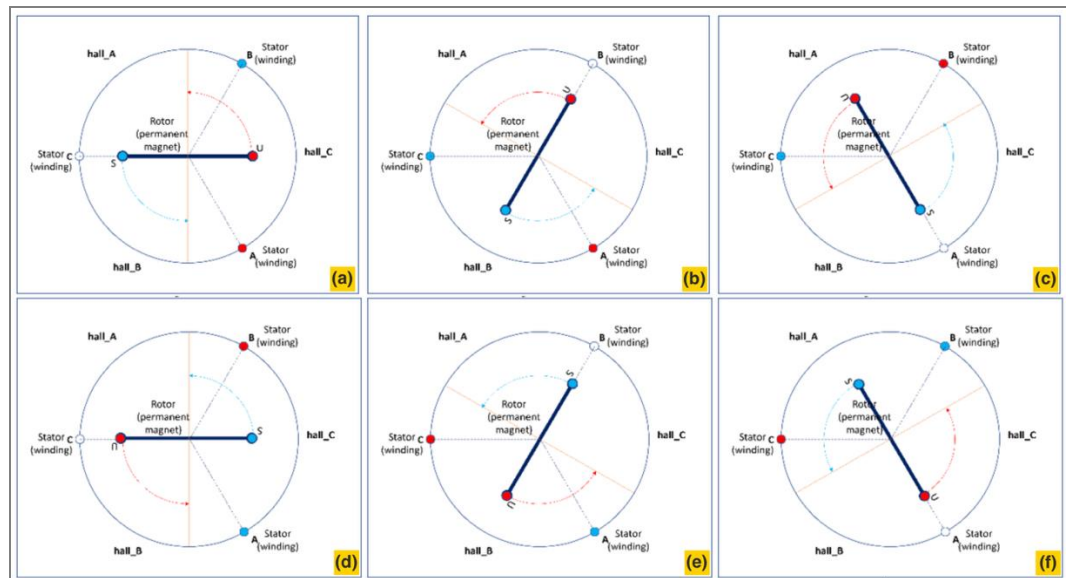


Figure 1.
Single pole BLDC
rotating sequence

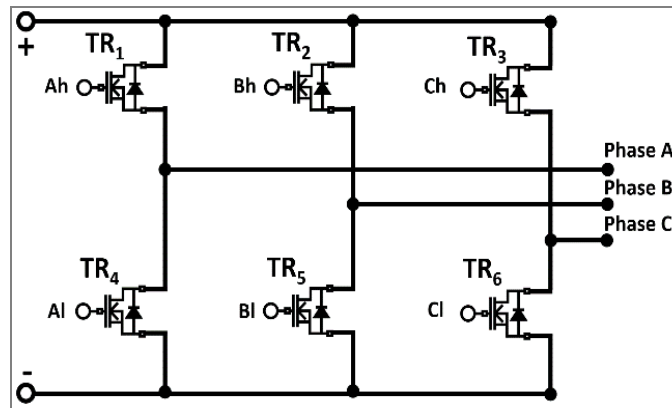
Table 1.
The sequence role of the
position, winding voltage,
hall sensor signal and the
sector

No	Position	Winding A / Phase A	Winding B / Phase B	Winding C / Phase C	hall_A	hall_B	hall_C	Sector
1	1a	$+V_{dc}$	$-V_{dc}$	NC	1	0	1	5
2	1b	$+V_{dc}$	NC	$-V_{dc}$	1	0	0	4
3	1c	NC	$+V_{dc}$	$-V_{dc}$	1	1	0	6
4	1d	$-V_{dc}$	$+V_{dc}$	NC	0	1	0	2
5	1e	$-V_{dc}$	NC	$+V_{dc}$	0	1	1	3
6	1f	NC	$-V_{dc}$	$+V_{dc}$	0	0	1	1

1.2. Three-Phase Inverter

Table 1 demonstrates that the voltage provided to the stator windings adheres to a defined commutation pattern to facilitate appropriate rotor movement. A three-phase inverter is employed

Figure 2.
Three-phase
inverter



to execute this switching sequence. **Figure 2** illustrates the schematic of the three-phase inverter circuit. The inverter functions as an electrical switch, functioning in a specified ON-OFF sequence to implement the commutation logic outlined in **Table 1**. The ON-OFF switching functionality is implemented utilizing power transistors, which function as programmable electronic switches. By supplying

a control current to the transistor's base terminal, the connection between the collector and emitter can be enabled or disabled. This control feature allows for accurate regulation of current flow to each winding phase.

In the commutation condition depicted in **Figure 1a**, to apply $+V_{dc}$ to phase A and $-V_{dc}$ to phase B, control signals are directed to transistors Ah (high-side switch for phase A) and Bl (low-side switch for phase B). In state 1_b, to apply $+V_{dc}$ to phase A and $-V_{dc}$ to phase C, the control signals are assigned to Ah and Cl, respectively. The procedure persists in states 1_c to 1_f, with the requisite combination of high-side and low-side transistors engaged for each phase. The comprehensive commutation switching logic governed by transistor control for all six phases of BLDC motor operation is encapsulated in **Table 2**. This table correlates each Hall sensor reading with the respective active transistors in the inverter, guaranteeing uniform and precise rotor motion during the cycle.

A cycle from 1a to 1f denotes a single electrical cycle, referred to as the *Six-Step* commutation mechanism. In a BLDC with a pair of poles, one electrical cycle corresponds to one mechanical cycle or a single rotor rotation (360°). If a BLDC motor has 4 pairs of poles, then one rotation of the rotor will correspond to 4 electrical cycles. A cycle from 1a to 1f denotes a single electrical cycle, referred to as the *Six-Step* commutation mechanism. In a BLDC with a pair of poles, one electrical cycle corresponds to one mechanical cycle or a single rotor rotation (360°). If a BLDC motor has 4 pairs of poles, then one rotation of the rotor will correspond to 4 electrical cycles.

Table 2.
The sequence role of the
position, phase voltage and
Transistor switching path

No	Position	Winding A / Phase A	Winding B / Phase B	Winding C / Phase C	Ah	Al	Bh	Bl	Ch	Cl
1	1a	$+V_{dc}$	$-V_{dc}$	NC	1	0	0	1	0	0
2	1b	$+V_{dc}$	NC	$-V_{dc}$	1	0	0	0	0	1
3	1c	NC	$+V_{dc}$	$-V_{dc}$	0	0	1	0	0	1
4	1d	$-V_{dc}$	$+V_{dc}$	NC	0	1	1	0	0	0
5	1e	$-V_{dc}$	NC	$+V_{dc}$	0	1	0	0	1	0
6	1f	NC	$-V_{dc}$	$+V_{dc}$	0	0	0	1	1	0

1.3. Six-Step Commutation of BLDC Motor

Integrating the principles of BLDC motor operating with a three-phase inverter enables effective motor rotation. This is achieved by delivering a suitable DC voltage to the inverter's input and employing rotor position data from Hall sensors to ascertain the correct switching sequence of the transistors within the inverter. The inverter's three-phase output is coupled to the motor windings, producing the necessary electromagnetic fields to commence and maintain rotor movement. Consequently, the BLDC motor operates at a rather stable speed determined by the commutation logic. Moreover, as depicted in **Figure 3**, motor speed can be modified by controlling the amplitude of the DC voltage sent to the inverter. This enables fundamental speed control capabilities; however, it currently lacks closed-loop feedback systems for dynamic regulation, such as those utilized by proportional-integral-derivative (PID) controllers.

PID controllers are widely used in many fields of engineering because they are easy to use, strong, and work well for real-time control. PID control has been successfully utilized in robotics to improve motion precision and system stability, as evidenced by quadrotor tracking applications employing visual feedback [15]. Other research shows that an adaptive PID control method used in the Robot Operating System (ROS) architecture makes quadcopter motion control far better than standard PID methods [16]. In electric vehicle (EV) applications, PID-based control has been

enhanced through the integration of fuzzy logic to improve robustness under nonlinear and varying operating conditions [17]. In renewable energy systems, PID control is effectively employed to enhance photovoltaic (PV) maximum power point tracking (MPPT) performance under varying irradiation and temperature conditions [18].

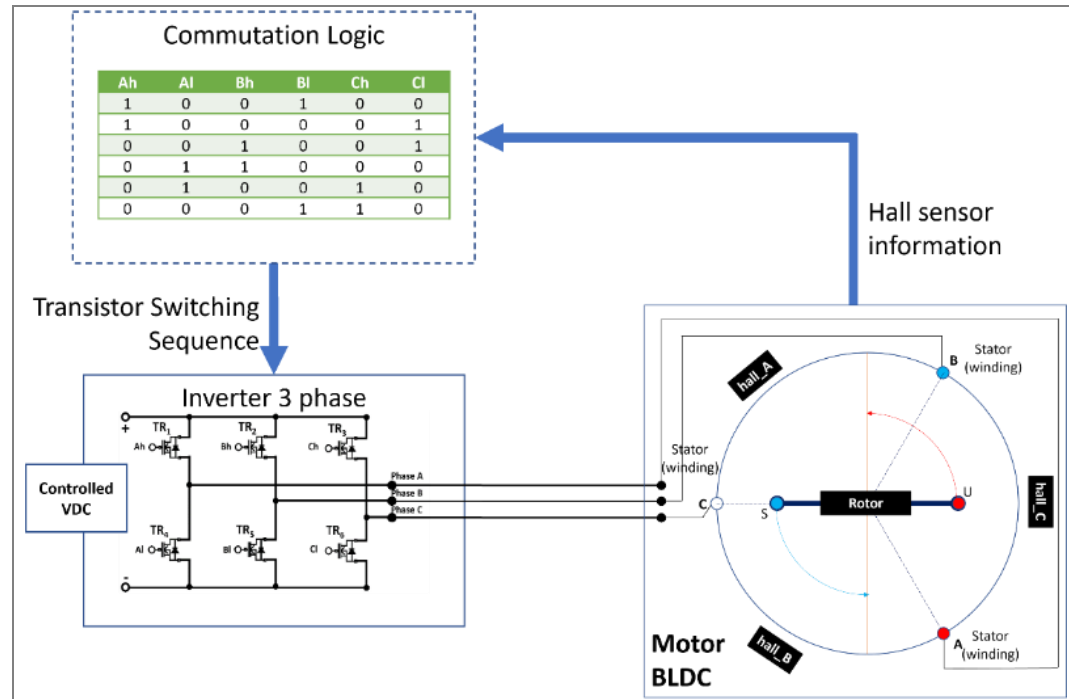


Figure 3.
Six-step commutation
implementation diagram

1.4. PID Control for BLDC Motor Speed

The setup depicted in Figure 3 exemplifies an open-loop control system, wherein the rotational speed of the BLDC motor is dictated exclusively by the amplitude of the DC voltage provided to the inverter. In an open-loop configuration, the motor speed is not compared to the required reference speed, so the system is unable to adjust to fluctuations in the mechanical load. A closed-loop control method is necessary to address this restriction. This method employs feedback from system output to modify input parameters, thereby augmenting output precision and bolstering resilience against external disturbances and load variations [19], [20].

This study specifically focuses on controlling the speed of the BLDC motor. Therefore, the key variable fed back into the system is the real-time motor speed. The control strategy employed is the Proportional–Integral–Derivative (PID) controller. This widely used technique combines three control actions—proportional, integral, and derivative—to minimize the error between the desired and actual speed. The mathematical representations of each control component are expressed in Eq. (1) for Proportional, Eq. (2) for Integral, and Eq. (3) for Derivative control.

$$P = K_p e(t) \quad (1)$$

$$I = K_i \int_0^t e(t) \quad (2)$$

$$D = K_d \frac{de(t)}{dt} \quad (3)$$

The control design scheme is illustrated in Figure 4. Let the reference speed be denoted as $r(t)$, and the motor speed, which serves as the system output, be $y(t)$. The error signal, defined as $e(t) = r(t) - y(t)$, is processed by the PID controller to generate a control signal $u(t)$. The resulting control law is formulated mathematically as shown in Eqs. (4) and (5), where $u(t)$ incorporates the proportional, integral, and derivative terms to regulate the system response.

$$u(t) = P + I + D \quad (4)$$

$$u(t) = K_p e(t) + K_i \int_0^t e(t) + K_d \frac{de(t)}{dt} \quad (5)$$

The selection of appropriate PID gain parameters—namely K_p , K_i , and K_d —can be achieved through various tuning techniques. In this study, the parameter tuning process is conducted using MATLAB's auto-tuning feature, which automatically adjusts the control gains based on the system's dynamic response to optimize performance and stability.

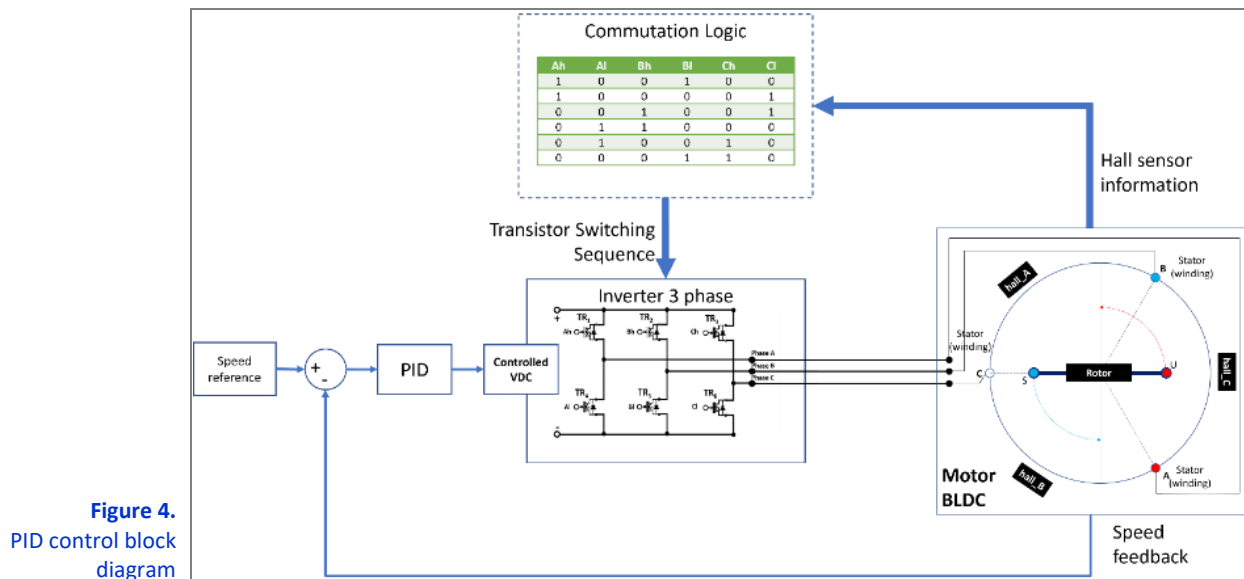


Figure 4.
PID control block
diagram

1.5. Matlab and Simulink

MATLAB, an abbreviation for "matrix laboratory," is a software platform created by MathWorks that facilitates matrix manipulation, function and data visualization, algorithm execution, user interface design, and integration with programs written in other programming languages [21]. It is extensively employed by researchers to model and examine many systems. This study employed MATLAB alongside its graphical extension, Simulink. Simulink is a MATLAB-based graphical programming platform for modeling, simulating, and evaluating multidomain dynamic systems. The main interface is a graphical block diagram tool, complemented with a customisable block library. Simulink provides seamless integration with the MATLAB environment and enables the execution of MATLAB scripts or embedded functions [22]. It is frequently utilized in domains such as automatic control systems and digital signal processing for multidomain simulation and model-based design.

The simulation in this study was developed using MATLAB 2020 with an educational license. Several block libraries provided by Simulink were utilized to construct the simulation model. Figure 5 presents a selection of the block libraries employed in this research.

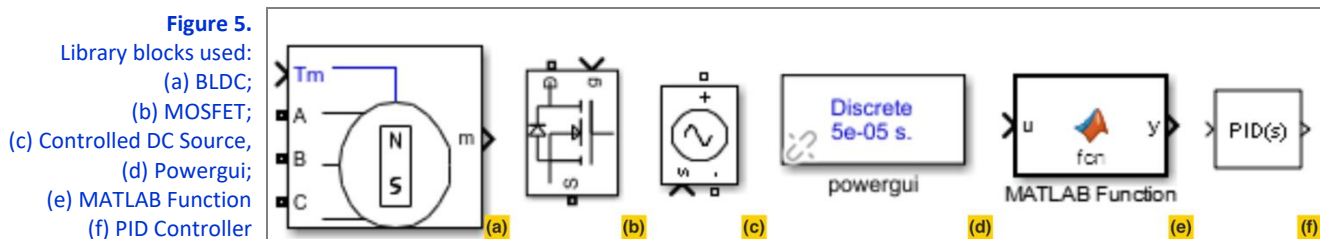


Figure 5.

Library blocks used:
(a) BLDC;
(b) MOSFET;
(c) Controlled DC Source,
(d) Powergui;
(e) MATLAB Function
(f) PID Controller

Figure 5 depicts the library blocks employed in the BLDC motor control simulation utilizing MATLAB Simulink. The BLDC block (Figure 5a) in the Simscape library denotes the Brushless DC motor model. This block is designed for a three-phase system featuring a trapezoidal back-EMF waveform. It comprises input ports for three phases (A, B, C) and one mechanical load input (T_m), in addition to an output port (m). The output signals, including hall sensor signals, rotor speed, electromagnetic torque, and back-EMF waveforms, are obtained via the Bus Selector block. A MOSFET block (Figure 5b) is employed for switching control in the three-phase inverter. This block comprises four terminals: drain, source, gate, and a measurement port, all of which can be deactivated if not required. The 160V DC voltage source is emulated using the Controlled Voltage Source block (Figure 5c), which produces voltage in response to an external input signal.

The Powergui block (Figure 5d) is utilized to specify the simulation type and sample period, which is established at 5×10^{-5} seconds for this investigation, in order to operate the system within a discrete environment. The Six-Step Commutation control logic is executed via the MATLAB Function block (Figure 5e), facilitating the incorporation of bespoke MATLAB code directly into the simulation framework. Finally, the PID Controller block (Figure 5f) is employed to implement closed-loop speed regulation, accommodating several control configurations such as PID, PI, PD, P-only, or I-only modes. This setup guarantees a precise, organized, and authentic simulation of BLDC motor action, successfully connecting theoretical models with practical application.

2. Methods

This study employed MATLAB/Simulink (R2022, educational license) to model and evaluate BLDC drive control through two complementary simulation models. First, an open-loop model was constructed to validate six-step electronic commutation against Hall-sensor states and the expected trapezoidal phase back-EMF; a three-phase inverter built from six MOSFET switches was driven by gate commands generated in a MATLAB Function that maps the Hall code (hall_A, hall_B, hall_C) to the A_h, A_i ; B_h, B_i ; C_h, C_i switching sequence. Second, a closed-loop speed-control model reused the same plant and commutation logic but routed measured rotor speed to a PID controller (auto-tuned in Simulink) to track time-varying speed references and reject load disturbances. Simulations were run in a discrete configuration (Powergui) with fixed sample time $\Delta t = 5 \times 10^{-5}$ s; the DC link was modeled as a 160 V controlled voltage source feeding the inverter and BLDC machine block. Each experiment lasted 8 s with the reference stepped $200 \rightarrow 400 \rightarrow 600 \rightarrow 800$ rpm at 1s intervals and then decreased to $700 \rightarrow 600 \rightarrow 500$ rpm at 1s intervals; at $t=7$ s a 10 Nm mechanical load step was applied while the reference was held at 500 rpm. Logged signals included rotor speed, three-phase stator currents (I_a, I_b, I_c), electromagnetic torque, Hall states, and phase back-EMF. The open-loop model confirmed that Hall codes, gate patterns, and back-EMF waveforms follow the theoretical six-sector sequence, while the closed-loop model quantified PID tracking and disturbance rejection under identical operating conditions. Together, these two models provide a clear and reproducible procedure for constructing, executing, and analyzing BLDC control simulations in Simulink.

3. Results and discussion

Two simulation models were built, with the first acting as the basis for the second. The initial model illustrates the simulation of the Six-Step Commutation process, which concurrently demonstrates the open-loop control of the BLDC motor. The second model improves upon the first by including a PID controller for closed-loop speed regulation of the BLDC motor.

3.1. Six-Step Commutation Simulation Model

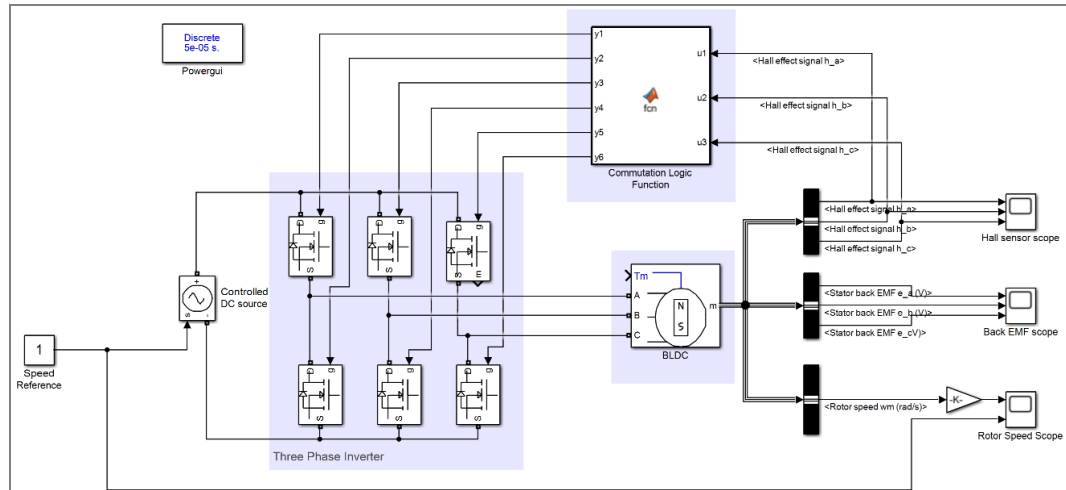
Figure 6 illustrates the open-loop simulation intended to exhibit the commutation process. Alongside the aforementioned library blocks seen in Figure 5, numerous ancillary blocks are incorporated, including Bus Selector, Scope, and Constant. Various scopes are employed to observe particular signals associated with commutation. The output signals from the BLDC motor are directed through the Bus Selector, which categorizes the signals into distinct scopes: the Hall signal scope for monitoring rotor position feedback, the Back-EMF scope for analyzing the waveform of the induced voltage, and the Switching signal scope for scrutinizing the gate signal sequence for each MOSFET. Additionally, a Speed scope is employed to juxtapose the actual speed of the BLDC motor with the reference speed designated by a Constant block. A Gain block is employed to convert motor speed from radians per second (rad/s) to revolutions per minute (RPM). This simulation facilitates meticulous observation of the switching pattern and rotor response, yielding an accurate assessment of the Six-Step Commutation process prior to the incorporation of any feedback control system.

In Figure 6, the regulated DC source is maintained at a constant value of 1 in this simulation. The illustration delineates the arrangement of the three-phase inverter circuit, consisting of six MOSFETs configured with two terminals linked to the DC voltage supply, six logic control inputs, and three-phase outputs connected to the BLDC motor. In this simulation, the T_m port of the BLDC motor is inactive, signifying the absence of any external mechanical demand. The outputs of the BLDC motor in the open-loop simulation comprise Hall sensor signals transmitted to the

commutation logic module, along with back-EMF and rotor speed signals, which are exhibited to monitor the motor's electrical and dynamic performance.

This study utilizes a MATLAB Function block featuring three input ports and six output ports to execute the commutation logic. The three input ports accept Hall sensor data from the BLDC motor, while the six output ports produce logic control signals for the MOSFETs in the three-phase inverter. The logical function integrated into the MATLAB Function block is formulated according to the conditions outlined in Table 1 and Table 2. This function dynamically establishes the correct switching sequence for the inverter transistors, guaranteeing accurate rotor movement in accordance with the *Six-Step* Commutation process.

Figure 6.
Open-loop simulation
circuit



```
function [y1,y2,y3,y4,y5,y6]= fcn(u1,u2, u3)
    input=[u1 u2 u3];
    y_out=zeros;
    if (input==[1 0 1])
        y_out=[1 0 0 1 0 0];
    else if (input==[1 0 0])
        y_out=[1 0 0 0 0 1];
    else if (input==[1 1 0])
        y_out=[0 0 1 0 0 1];
    else if (input==[0 1 0])
        y_out=[0 1 1 0 0 0];
    else if (input==[0 1 1])
        y_out=[0 1 0 0 1 0];
    else if (input==[0 0 1])
        y_out=[0 0 0 1 1 0];
    end
    end
    end
    end
    end
    end
    end
    end
    y1 = y_out(1,1);
    y2 = y_out(1,2);
    y3 = y_out(1,3);
    y4 = y_out(1,4);
    y5 = y_out(1,5);
    y6 = y_out(1,6);
```

3.2. Six-Step Commutation Simulation Results

In this work, the open-loop simulation circuit was executed with a total simulation time of 3 seconds. The simulation output is illustrated in Figure 7. Figure 7a illustrates that the sequence of Hall sensor signals produced throughout the simulation aligns exactly with the commutation pattern outlined in Table 1. Figure 7b illustrates the produced back-EMF waveform, which distinctly displays a trapezoidal form, consistent with the anticipated profile for a BLDC motor, as indicated

in Table 1. The commutation logic's switching sequence (Figure 7c) adheres to the Six-Step commutation pattern outlined in Table 2.

The rotation of the motor is evidenced by the speed monitoring scope depicted in Figure 8. The motor commences at 0 RPM and progressively accelerates until attaining a stable speed. The illustration distinctly indicates that the speed reference is established at 1 RPM, whereas the actual speed escalates from 0 to roughly 10 RPM. The divergence between the reference and actual speed arises from the open-loop configuration of the simulation and the lack of mechanical load on the motor. A closed-loop control system is necessary to enhance alignment between the reference speed and the actual motor speed. This feature will be detailed in the following part, which examines the simulation results of the BLDC motor utilizing PID-based speed control.

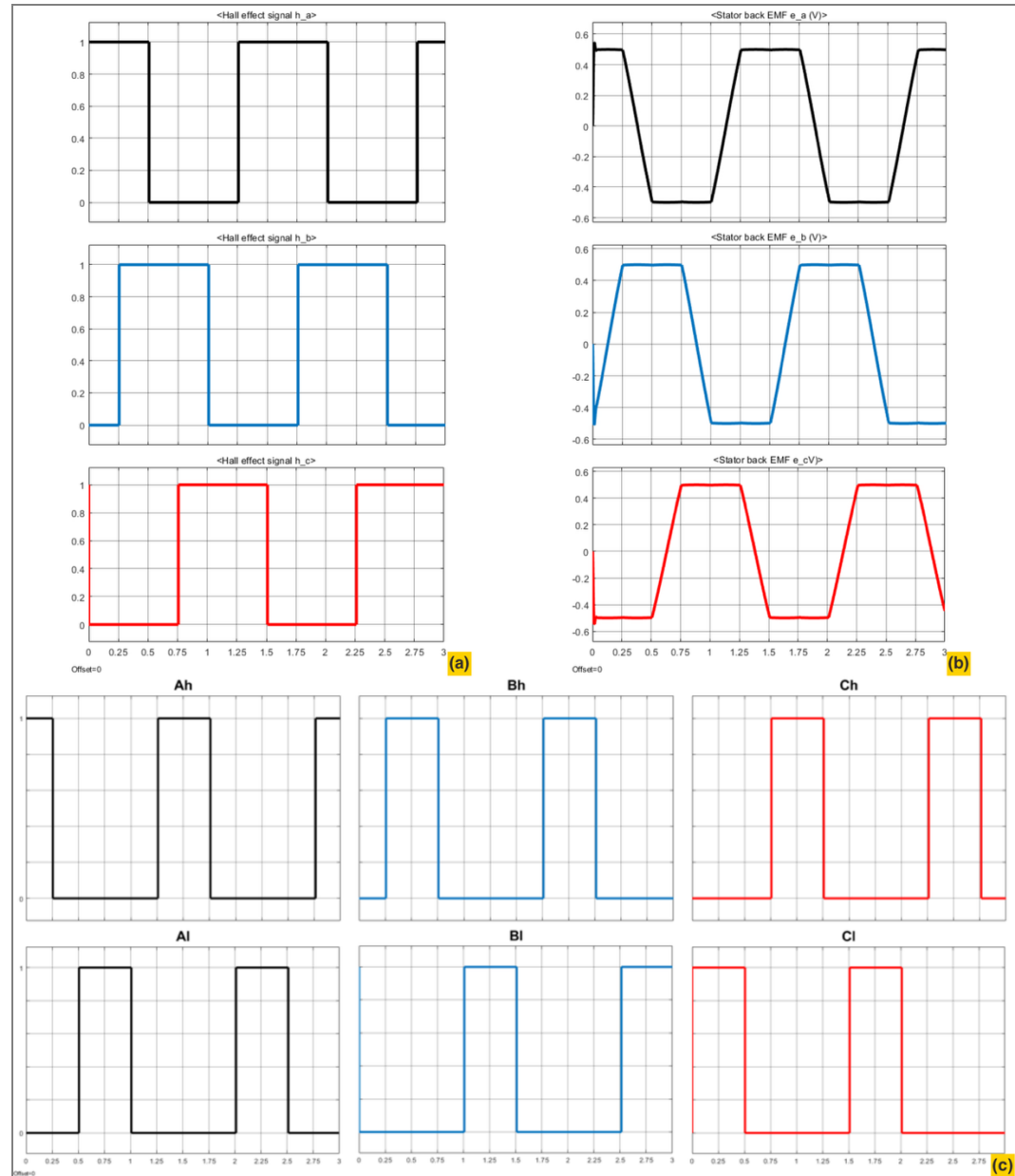


Figure 7.
Simulation results for:
(a) Hall sensor data;
(b) Back-EMF;
(c) Switching data

3.3. Simulation of BLDC Motor Speed Control Using PID Method

The simulation circuit for BLDC motor speed regulation via the PID approach is an improvement over the previously illustrated open-loop simulation depicted in Figure 6. The principal advancement introduced at this stage is the incorporation of a PID controller block. Figure 9 illustrates the use of a Signal Builder to establish the reference speed profile, while a Step function is employed to replicate the application of load to the BLDC motor. Modifications to the output configuration from the Bus Selector to the Scopes were implemented to enhance the data analysis process. The complete simulation can be download from this link: <https://bit.ly/bldcsimulation>.

Figure 8.
Speed data, speed
reference (red), and
actual speed (black)

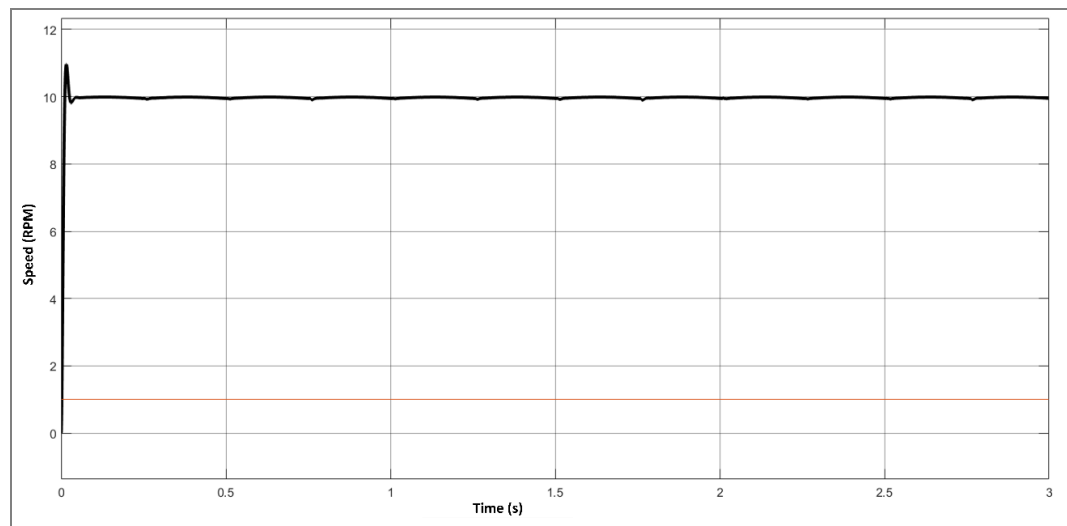
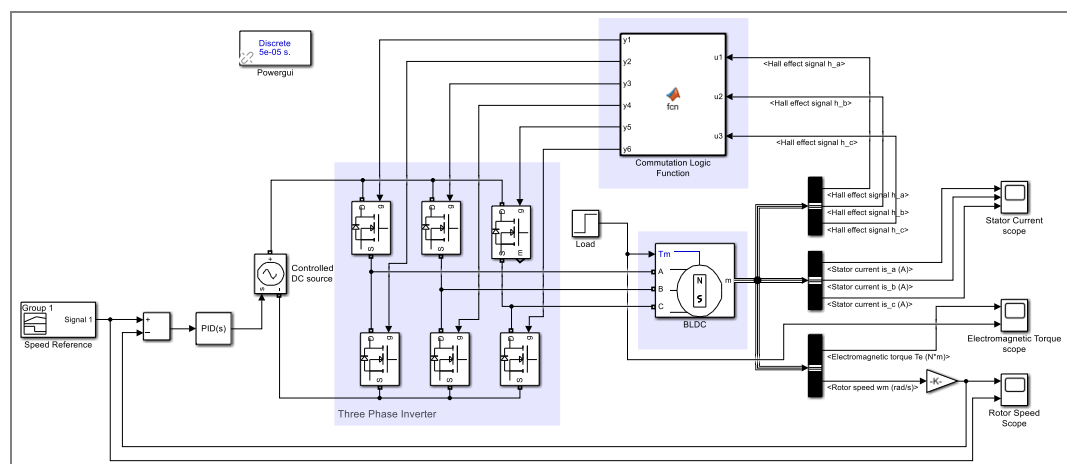


Figure 9.
Closed-loop BLDC
speed control simulation
circuit



The simulation results will display the rotor speed reaction through the speed scope, the phase current response through the stator current scope, and the electromagnetic torque response through the torque scope. The simulation was conducted over 8 seconds, involving fluctuations in the reference speed (both increasing and decreasing) and the application of load to assess system robustness and controller efficacy.

3.4. Simulation Results of BLDC Motor Speed Control using the PID Method

Figure 10 illustrates the motor response under varying speed references and load conditions. From 0 to 4 seconds, the speed reference is gradually increased every second from 200 RPM to 400 RPM, 600 RPM, and 800 RPM. Subsequently, the reference is decreased to 700 RPM, 600 RPM, and 500 RPM in 1-second intervals. At the 7th second, a load of 10 Nm is applied while the reference speed remains at 500 RPM.

Figure 10 depicts the simulation outcomes of the BLDC motor speed regulation employing the PID methodology under dynamic situations. Between 0 and 4 seconds, the speed reference is incrementally raised from 200 RPM to 800 RPM, thereafter reduced to 700 RPM, 600 RPM, and 500 RPM at 1-second intervals. At the 7-second interval, a mechanical stress of 10 Nm is applied while the reference speed is sustained at 500 RPM. Additionally, **Figure 10a** illustrates the rotor speed response, wherein the real speed (blue line) closely adheres to the reference (purple line) with negligible delay. Despite minimal overshoot and undershoot during transitions, the motor promptly stabilizes following each alteration. Upon application of load, the motor demonstrates a temporary underspeed before stabilizing at the desired velocity. **Figure 10b** illustrates the current response over the three stator phases (I_a , I_b , and I_c), thereby affirming the motor's three-phase configuration. The current magnitude escalates with velocity and increases markedly upon the introduction of the load. Moreover, **Figure 10c** illustrates the electromagnetic torque response, which escalates in direct correlation with speed and experiences additional spikes upon load

application. Control-oriented interpretation. The phase currents increase with reference speed and exhibit a transient rise after the 10 Nm load step, coinciding with a torque spike followed by stabilization. These profiles explain the brief speed undershoot and the integral term's corrective effect. From a control standpoint, the current and torque traces provide actionable cues for PID refinement: (i) limit integral windup during load transients, (ii) adjust the proportional gain to reduce torque ripple that appears as speed ripple near commutation, and (iii) consider soft-commutation timing to mitigate current spikes (di/dt).

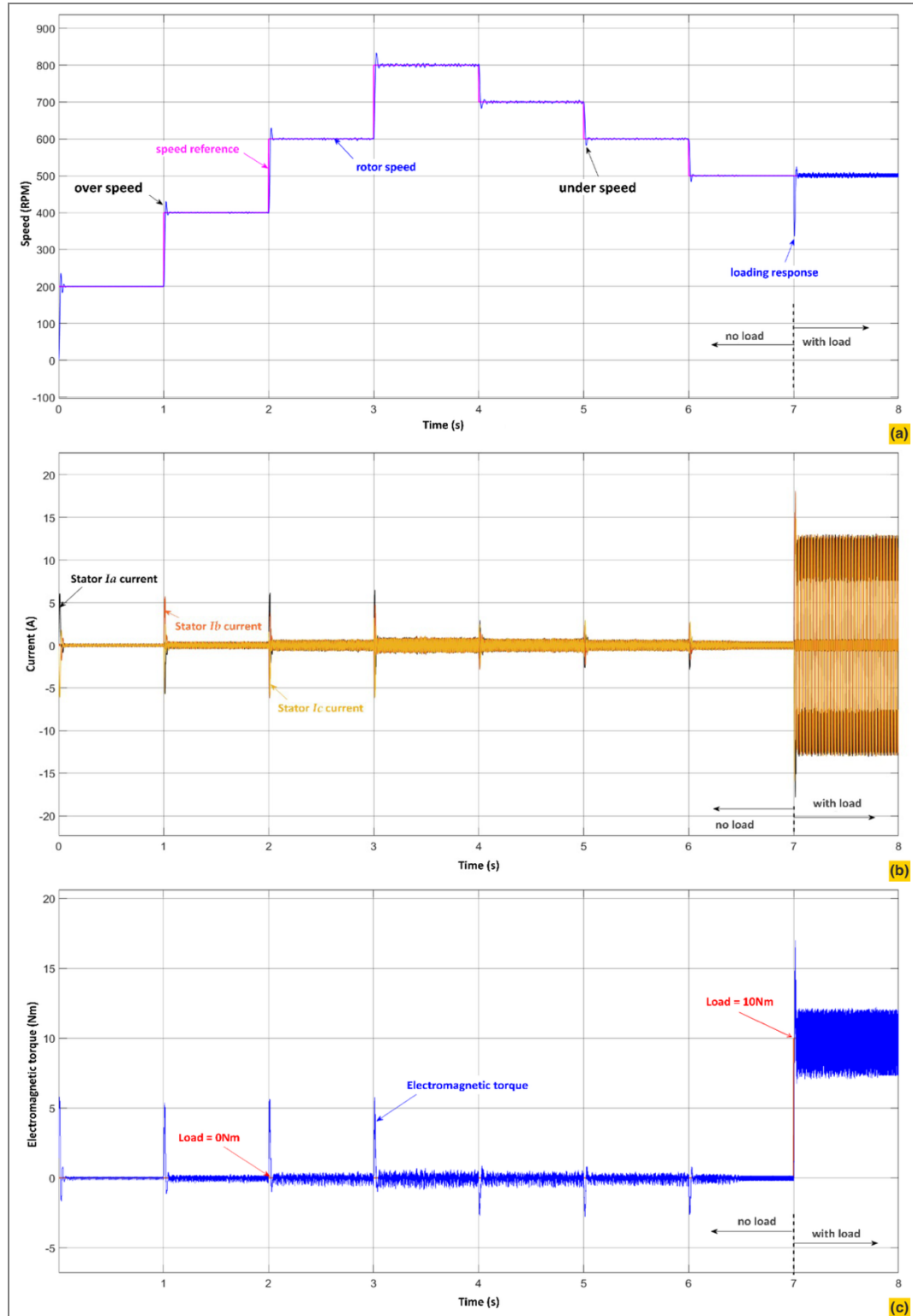


Figure 10.
BLDC speed control
simulation results with
PID

The simulation findings validate the efficacy of the PID-based speed control in ensuring stable operation under diverse settings. The controller effectively monitors the reference speed and mitigates load disturbances, showcasing strong system performance. The simulation verifies the

accurate execution of *Six-Step* commutation, confirming appropriate switching logic and rotor alignment.

This modeling method substantially mitigates the chance of hardware failure during physical implementation by facilitating comprehensive testing and analysis in advance. Notwithstanding these results, the simulations employ idealized elements; ideal switching devices (no dead-time or switching losses), a stiff 160 V DC link, perfectly balanced phases, fixed motor parameters (e.g., K_e , K_t , inertia), and ideal Hall-sensor signals under a discrete $\Delta t = 5 \times 10^{-5}$ s configuration. In hardware, nonidealities such as device on-resistance and diode reverse-recovery, dead-time induced commutation intervals, DC-link ripple and wiring inductance, sensor quantization/latency and misalignment, as well as temperature-dependent parameter drift, can introduce additional speed/torque ripple and slightly lengthen settling following load steps. Consequently, hardware performance may show modest increases in overshoot or steady-state error relative to the simulation baseline (where we observed $\approx \pm 2\%$ speed regulation). To bridge this gap in our planned experimental work, we will incorporate dead-time insertion, current limiting with anti-windup, soft-commutation timing to reduce di/dt , appropriate DC-link capacitance and snubber sizing, and parameter identification for K_e/K_t and inertia, and then compare logged responses against the present Simulink models.

Moreover, employing MATLAB Simulink's auto-tuning capability enables optimal parameter selection for the PID controller, hence improving control precision. These findings underscore the utility of simulation tools for both validation and performance forecasting in real-world scenarios, providing a basis for the future advancement of more sophisticated BLDC control techniques.

In addition to helping visualize the response of the control system, the successful simulation in this paper allows several opportunities for further development. The simulation process allows for maximum loading on the system without incurring risk. The simulation process can also allow the testing of various parameter variations to find the best solution and become a platform for continuous system improvement. Some of the development opportunities that will be carried out include trying to develop new algorithms and of course realizing the experimental hardware implementation.

Beyond validating the control scheme, the simulation framework enables further optimization through parameter sensitivity analysis, tuning refinement under nonlinear loads, and exploration of advanced control algorithms such as fuzzy-PID and model predictive control (MPC). These enhancements, alongside hardware validation, constitute the next phase of this ongoing research.

4. Conclusion

This research effectively established a foundational simulation model for BLDC motor control with Simulink-MATLAB. The simulation occurred in two phases: an open-loop simulation to verify the *Six-Step* commutation procedure, and a closed-loop simulation employing PID control to manage motor speed. The *Six-Step* commutation logic was successfully executed using a MATLAB Function block, and its accuracy was confirmed by analyzing the hall signals, back-EMF waveforms, and switching signal sequences. The PID control system effectively regulated the motor speed in accordance with reference commands, as evidenced by speed response analysis. The simulation offered insights into the phase current and electromagnetic torque under different speed references and applied loads. The simulation offered insights into the phase current and electromagnetic torque under different speed references and applied loads. Results indicate that elevated speed and applied torque correlate with increased current and electromagnetic torque values. These current and torque behaviors clarify the controller's compensation dynamics under load disturbances and directly inform control refinement: targeted PID retuning to reduce speed overshoot/undershoot, anti-windup action to prevent prolonged recovery after load steps, and soft-commutation and current-limiting strategies to mitigate torque ripple and device stress. Consequently, the simulation results not only validate the present scheme but also delineate concrete avenues for improving drive smoothness and efficiency in future BLDC implementations.

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Authors' Declaration

Authors' contributions and responsibilities - The authors made substantial contributions to the conception and design of the study. The authors took responsibility for data analysis, interpretation and discussion of results. The authors read and approved the final manuscript.

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